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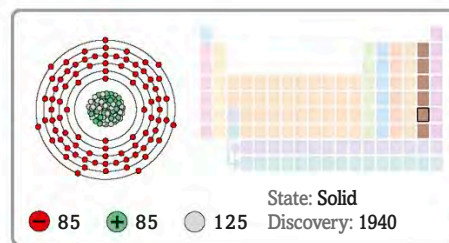
Astatine

Atoms of astatine are unstable, and typically break down after just a few hours, into atoms of lighter elements, such as **bismuth**. This radioactive element itself forms in a similar way when atoms of a heavier element called francium break apart. Tiny amounts of this rare element are found in uranium ores, such as **uraninite**. The Italian physicist Emilio Segrè was one of the first scientists to isolate a sample of pure astatine. He was able to do so by using a particle accelerator: this is a machine that smashes together atoms and then studies the results.



Uraninite

Inside this mineral, unstable atoms of the element francium are breaking apart, forming astatine atoms.



The Halogen Group

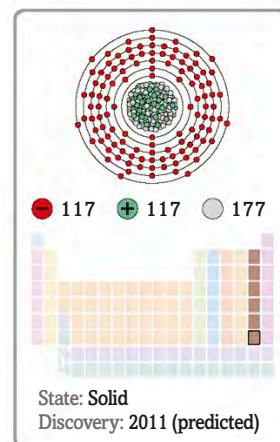
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Tennessine

Nuclear Reactor, Oak Ridge National Laboratory, Tennessee, USA



Atoms of tennessine existed for a few seconds after they were formed.



Tennessine is the youngest element in the periodic table. It was produced in 2011, in the Russian city of Dubna. The element was named after the US state of Tennessee, home to the **Oak Ridge National Laboratory**, which houses one of the first, large-scale nuclear reactors ever built. Only a few atoms of this halogen element have ever been made. Even so, scientists have predicted it to be a semi-metal, not a non-metal like all the other halogens.